

Chung-Yen (Matt) Chiang

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Research Interests

Network systems — high-performance datapath, kernel-bypass and offload, and cross-layer performance measurement.

Education

Seoul National University

2026 – 2028

M.S. in Computer Science and Engineering

Advisor: Prof. KyoungSoo Park

National Taiwan University of Science and Technology

2019 – 2023

B.B.A. in Information Management

GPA: 4.01/4.30

Publications

[C1] **C.-Y. Chiang** and N.-W. Lo. “Secure Channels for Smart Manufacturing Systems in Cloud Environments,” in *Proceedings of the Cryptology and Information Security Conference (CISC)*, 2022. (1st author)

Research Experience

T-Networking Laboratory, Seoul National University

Sep 2026 – Present (incoming)

Graduate Researcher (Advisor: Prof. KyoungSoo Park)

- Joining Prof. Park’s group to conduct research in network systems; specific research direction to be determined upon program start.

Institute of Information Science, Academia Sinica

Jul 2022 – Aug 2022

Summer Research Intern (Advisor: Prof. Meng Chang Chen)

- Researched hardware-level sandbox evasion techniques (CPU feature checks, registry artifacts) used by obfuscated malware to detect analysis environments, aiming to improve sandbox detection coverage.
- Reverse-engineered obfuscated malware samples using **IDA Pro** to identify evasion triggers and anti-analysis behaviors.
- Implemented a modular detection framework in C and Python extending the **Cuckoo Sandbox** monitor with new evasion-detection plugins.

Secure Channels for Smart Manufacturing Systems in Cloud Environments *Jul 2021 – Feb 2022*

Undergraduate Researcher (Advisor: Prof. Nai-Wei Lo)

- Designed and implemented **Site-to-Site IPsec** secure channels for industrial communication between local gateways and AWS VPC; published as 1st author at CISC 2022.
- Conducted deep packet inspection with **Wireshark** and benchmarked throughput/latency with **iPerf3** and **Netperf** across multiple IPsec configurations.
- Investigated **ARMv8 Cryptographic Extensions** to accelerate encryption on resource-constrained industrial devices.

Industry Experience

Amazon - Connectivity Team

Apr 2026 – Jun 2026

Software Development Engineer II (L5)

- Conducted **cross-layer performance profiling** of an end-to-end Wi-Fi data path on **FreeRTOS**, spanning PHY, SDIO, kernel, and application layers; identified IMI memory access (SDIO), TCP shared-lock and mutex contention (kernel), and application-layer memory contention as root-cause bottlenecks, achieving **2.36× throughput improvement** (10.59 to 25 Mbps).
- Designed runtime **dynamic NVRAM/CLM loading** mechanism for multi-SKU and multi-region firmware deployment from a unified image.

Amazon - Connectivity Team*Aug 2023 – Mar 2026***Software Development Engineer I (L4)**

- Characterized DHCP reconnection behavior under unstable network conditions and designed protocol-level optimizations, reducing reconnection latency by **76%**.
- Conducted experimental tuning of **WMM/EDCA QoS** parameters under traffic contention, improving p99 latency for latency-sensitive streaming by **24%**.
- Implemented Wi-Fi power-saving and wake-up mechanisms, extending battery-operated IoT device standby from **40 to 65 days**.
- Built a modular BLE provisioning subsystem with end-to-end encryption channel for global SKU deployment.

Amazon - Connectivity Team*Sep 2022 – Jul 2023***Software Development Engineer Intern**

- Developed automated tooling for interoperability characterization across **60+ commercial Wi-Fi routers**, building a dataset for cross-vendor connection behavior analysis.

Teaching Experience

Cybersecurity and AI-based Analytics*Fall 2022***Teaching Assistant, Dept. of Electrical Engineering, NTUST**

- Supported instruction of a graduate course covering network security fundamentals, threat detection, and machine-learning-based security analytics.
- Held weekly office hours, graded assignments and exams, and provided guidance for hands-on lab exercises and final projects.

Honors & Awards

MOST Undergraduate Research Scholarship*2021**Ministry of Science and Technology, Taiwan***Academic Excellence Award***2021 – 2022**NTUST — Top 2 of class for three consecutive semesters***Technical Skills**

Systems	Embedded Systems, Wi-Fi & BLE, RTOS (FreeRTOS), Cross-Layer Performance Profiling, BSP/SDK Integration
Programming	C, C++, Python, Shell Scripting
Tools	Linux, GDB / JTAG, perf, ftrace, Wireshark, tcpdump, iperf3, Git
Languages	English (Professional), Mandarin (Native)